📘 **AI/ML Internship – Day 40 Notes & Tasks**

🟢 **Module 7: Decision Tree Classification Algorithm**

📌 **Part 1: Introduction to Decision Trees**

🔹 **What is a Decision Tree?**

👉 A Decision Tree is a Supervised Machine Learning algorithm used for:

- Classification
- Regression

👉 It makes decisions by asking a series of questions.

🔹 **Why is it called a Decision Tree?**

Because it looks like an upside-down tree:

```
        Is Study Hours > 4?
               |
     ------------------
     |              |
    Yes            No
     |              |
   Pass           Fail
```

The model reaches a decision by following branches.

📌 **Part 2: Real-Life Example**

Imagine a teacher wants to predict whether a student will pass.

Questions:

Study Hours > 4 ?

Yes → Pass

No → Fail

This question-based decision making is exactly how Decision Trees work.

📌 Part 3: Theory Answers (IMPORTANT)

🔹 1. What is a Decision Tree?

👉 A Decision Tree is a supervised learning algorithm that makes predictions using decision rules.

🔹 2. Is Decision Tree Used for Classification or Regression?

👉 It can be used for both classification and regression.

🔹 3. What is the Root Node?

👉 The first question in the tree.

Example:

Is Study Hours > 4?

🔹 4. What is a Branch?

👉 A path connecting one decision to another.

🔹 5. What is a Leaf Node?

👉 The final prediction/output.

Example:

Pass

Fail

📌 Part 4: Important Terms

Term	Meaning
Root Node	Starting point

Term	Meaning
Branch	Decision path
Leaf Node	Final answer
Split	Dividing data
Tree Depth	Levels in tree

📌 Part 5: Example Dataset

Study Hours	Result
1	Fail
2	Fail
3	Fail
5	Pass
6	Pass
7	Pass

📌 Part 6: Visual Tree Representation

```
       Study Hours > 4 ?
              |
     ---------------------
     |              |
    Yes             No
     |              |
    Pass            Fail
```

📌 Part 7: Install Required Library

pip install scikit-learn

📌 Part 8: Import Libraries

import pandas as pd

from sklearn.tree import DecisionTreeClassifier

📌 Part 9: Create Dataset

```
data = {
   "Hours":[1,2,3,5,6,7],
   "Result":[0,0,0,1,1,1]
}

df = pd.DataFrame(data)

print(df)
```

Meaning

Value	Meaning
0	Fail
1	Pass

📌 Part 10: Features and Labels

```
X = df[["Hours"]]

y = df["Result"]
```

👉 X = Feature

👉 y = Label

📌 Part 11: Create Decision Tree Model

```
model = DecisionTreeClassifier()
```

📌 Part 12: Train the Model

model.fit(X, y)

👉 The model learns decision rules from data.

📌 **Part 13: Make Prediction**

prediction = model.predict([[4]])

print(prediction)

Possible Output:

[0]

Meaning:

👉 FAIL

📌 **Part 14: Predict Multiple Values**

prediction = model.predict([[2],[5],[8]])

print(prediction)

Possible Output:

[0 1 1]

📌 **Part 15: Complete Program**

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier

data = {
   "Hours":[1,2,3,5,6,7],
   "Result":[0,0,0,1,1,1]
}
```

```
df = pd.DataFrame(data)

X = df[["Hours"]]
y = df["Result"]

model = DecisionTreeClassifier()

model.fit(X, y)

prediction = model.predict([[5]])

print("Prediction:", prediction)
```

📌 Part 16: Advantages of Decision Tree

✅ Easy to understand

✅ Easy visualization

✅ Handles classification and regression

✅ No complex mathematics

📌 Part 17: Disadvantages of Decision Tree

❌ Can overfit data

❌ Sensitive to small data changes

❌ Large trees become complex

📌 Part 18: Real-World Applications

Education

👉 Student Performance Prediction

Banking

👉 Loan Approval Prediction

Healthcare

👉 Disease Prediction

E-Commerce

👉 Customer Classification

Security

👉 Fraud Detection

📌 Part 19: Decision Tree vs KNN

Decision Tree	KNN
Learns decision rules	Uses nearest neighbors
Fast prediction	Slower prediction
Easy visualization	No tree structure
Good interpretability	Less interpretable

📌 Part 20: Overfitting in Decision Trees

🔹 What is Overfitting?

👉 When a model memorizes training data instead of learning patterns.

Result:

- Excellent training performance
- Poor performance on new data

Solution

Limit tree depth:

```
model = DecisionTreeClassifier(max_depth=3)
```

📌 Part 21: Decision Tree Workflow

Dataset
↓
Features (X)
↓
Decision Rules
↓
Tree Creation
↓
Prediction

🎯 Tasks for Day 40

🧠 Task 1: Theory

1. What is a Decision Tree?
2. What is a Root Node?
3. What is a Branch?
4. What is a Leaf Node?
5. Advantages of Decision Trees?

💻 Task 2: Dataset Creation

1. Create student result dataset
2. Create loan approval dataset
3. Create disease prediction dataset

💻 Task 3: Model Building

1. Create Decision Tree model
2. Train model
3. Make predictions

💻 Task 4: Prediction Practice

Predict:

1. Student result
2. Disease status
3. Loan approval

💻 Task 5: Tree Analysis

1. Identify root node
2. Identify leaf nodes
3. Explain decision path

💻 Task 6: Compare Algorithms

Compare:

1. Logistic Regression
2. KNN
3. Decision Tree